A DEGREE-THREE SPECTRAL REFINEMENT OF NISHIOKA'S LOWER BOUND FOR THE THREE-DIMENSIONAL BLASCHKE–LEBESGUE PROBLEM

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ABSTRACT. Let $K \subset \mathbb{R}^3$ be a convex body of constant width d. Nishioka proved $\mathrm{Vol}(K) \geq (4\pi/33)d^3$ by combining a support-function formulation with a spherical-harmonic estimate. We retain the degree-three harmonic component of the centered support function as an additional variable and derive a sharp quartic determinant inequality for its curvature tensor. This gives the candidate refinement

$$\mathrm{Vol}(K) \geq \frac{\pi}{1914}\left(259 - 27\sqrt{\frac{15183}{17303}}\right) d^3 = 0.3836027047090677\ldots\, d^3.$$

The central tensor identity is derived invariantly and is also checked by two exact symbolic implementations, exhaustive pairing enumeration, and independent Python and R numerical tests. This is an AI-assisted research draft seeking independent mathematical verification. A recent public HYRA manuscript claims substantially stronger bounds by a different geometric and computer-assisted method; the contribution proposed here is an independent spectral mechanism rather than the numerically strongest current claim.

Verification status. This manuscript and its accompanying code form a non-peer-reviewed research draft. Exact computer algebra supports the displayed tensor identities, but no independent subject-matter expert has yet reviewed the complete proof. The repository should therefore be cited as a draft seeking verification, not as a certified or peer-reviewed theorem.

1. Introduction

The three-dimensional Blaschke–Lebesgue problem asks for the least possible volume of a convex body of prescribed constant width. The conjectured minimizers are the two Meissner tetrahedra, whose common volume is approximately $0.419860046d^3$; the problem remains open [4, 6].

Improving Chakerian's classical estimate [1], Nishioka recently established

$$\mathrm{Vol}(K) \geq \frac{4\pi}{33}d^3 = 0.380799109526\ldots\, d^3$$

using the centered support function, spherical harmonics, and Bochner's formula [6]. His discussion identifies a source of slack: equality in the degree-three spectral estimate is incompatible with equality in the pointwise curvature-matrix bound. He suggests going beyond the rotation-averaged relaxation by introducing additional variables or symmetry-breaking information.

The present note retains the degree-three component h_3 of the centered support function and estimates the curvature tensor A_{h_3} more sharply than one can estimate an arbitrary tangent-matrix field. The key new input is the sharp inequality

$$\|4 \det A_Y\|^2_{L^2(\mathbb{S}^2)} \leq \frac{15183}{17303\pi} \|A_Y\|^4_{L^2(\mathbb{S}^2)}, \qquad Y \in \mathcal{H}_3. \tag{1}$$

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It quantifies a restriction intrinsic to cubic spherical harmonics and leads to a reduced degree-three curvature budget.

Harrell's variational formulation in terms of the sum of principal radii of curvature supplies useful historical and conceptual context: in dimension three that formulation is a relaxation because not every admissible scalar curvature field comes from an embedded constant-width body [2, 6]. The argument here instead remains within Nishioka's exact support-function constraints.

A public HYRA manuscript dated August 2026 claims the stronger analytic bound $0.4041207788d^3$ and a computer-assisted bound above $0.4110404737d^3$ [3]. Those claims have not been checked in this project. If correct, they supersede the numerical coefficient obtained here. The purpose of this note is therefore to document a complementary spectral route in a transparent, reproducible form.

2. Support functions and the refined spectral split

We follow Nishioka's normalization [6]. Let p be the support function of a smooth body of constant width d and put

$$h = p - \frac{d}{2}.$$

Then $h(-u) = -h(u)$. After a translation we may also impose $h \perp \mathcal{H}_1$. Define the self-adjoint tangent endomorphism

$$A_h := \nabla^2 h + h\,\mathrm{Id}\,.$$

The principal radii of curvature of p are the eigenvalues of $A_h + (d/2)\,\mathrm{Id}$, and constant width gives the pointwise bounds

$$-\frac{d}{2}\,\mathrm{Id} \preceq A_h(u) \preceq \frac{d}{2}\,\mathrm{Id}, \qquad u \in \mathbb{S}^2. \tag{2}$$

Moreover,

$$\mathrm{Vol}(K) = \frac{\pi d^3}{6} - \frac{d}{2}E(h), \qquad E(h) := \int_{\mathbb{S}^2} \left(\frac{1}{2}\,|\nabla h|^2 - h^2 \right) d\sigma. \tag{3}$$

For tangent endomorphism fields A, B, write

$$\langle A, B \rangle_2 := \int_{\mathbb{S}^2} \mathrm{tr}(AB)\,d\sigma, \qquad \|A\|_2^2 := \langle A, A \rangle_2.$$

If $Y_\ell \in \mathcal{H}_\ell$ and $\lambda_\ell = \ell(\ell+1)$, Nishioka's calculation gives

$$E(Y_\ell) = \frac{\lambda_\ell - 2}{2}\,\|Y_\ell\|_{L^2}^2\,, \qquad \|A_{Y_\ell}\|_2^2 = (\lambda_\ell - 1)(\lambda_\ell - 2)\,\|Y_\ell\|_{L^2}^2\,. \tag{4}$$

We first record explicitly the tensor-valued orthogonality needed below.

Lemma 2.1 (Polarized Bochner identity)**.** *For smooth real-valued f, g on $\mathbb{S}^2$,*

$$\int_{\mathbb{S}^2} \left\langle \nabla^2 f, \nabla^2 g \right\rangle d\sigma = \int_{\mathbb{S}^2} (\Delta f)(\Delta g)\,d\sigma - \int_{\mathbb{S}^2} \langle \nabla f, \nabla g \rangle\; d\sigma.$$

Proof. Polarize the integrated Bochner identity on the unit sphere,

$$\int_{\mathbb{S}^2} \left|\nabla^2 q\right|^2 d\sigma = \int_{\mathbb{S}^2} (\Delta q)^2\,d\sigma - \int_{\mathbb{S}^2} |\nabla q|^2\; d\sigma,$$

by applying it to $q = f + g$ and $q = f - g$. □

Lemma 2.2 (Curvature-tensor orthogonality)**.** *If $f \in \mathcal{H}_\ell$ and $g \in \mathcal{H}_m$ with $\ell \neq m$, then $\langle A_f, A_g \rangle_2 = 0$. If $f, g \in \mathcal{H}_\ell$, then*

$$\langle A_f, A_g \rangle_2 = (\lambda_\ell - 1)(\lambda_\ell - 2) \int_{\mathbb{S}^2} f g\,d\sigma.$$

Proof. Expanding and then using Lemma 2.1,

$$\begin{aligned}\langle A_f, A_g\rangle_2 = &\int_{\mathbb{S}^2}(\Delta f)(\Delta g)\,d\sigma - \int_{\mathbb{S}^2}\langle \nabla f, \nabla g\rangle\,d\sigma\\ &+\int_{\mathbb{S}^2} g\Delta f\,d\sigma + \int_{\mathbb{S}^2} f\Delta g\,d\sigma + 2\int_{\mathbb{S}^2} fg\,d\sigma.\end{aligned}$$

Distinct harmonic degrees are L^2-orthogonal, and every term above is a scalar multiple of $\int fg$. For equal degrees, substitute $\Delta f = -\lambda_\ell f$ and $\Delta g = -\lambda_\ell g$ to obtain the stated factor. □

Write the odd harmonic expansion as

$$h = h_3 + h_{\geq 5}, \qquad h_{\geq 5} = \sum_{\substack{\ell\geq 5\\ \ell \text{ odd}}} h_\ell,$$

and put

$$C := A_{h_3}, \qquad N := \|A_h\|_2^2, \qquad N_3 := \|C\|_2^2.$$

Since h is smooth, its harmonic expansion converges in $H^2(\mathbb{S}^2)$; hence Lemma 2.2 gives

$$N = \sum_{\substack{\ell\geq 3\\ \ell \text{ odd}}} \|A_{h_\ell}\|_2^2, \qquad N_3 = \langle A_h, C\rangle_2. \tag{5}$$

Proposition 2.3 (Refined spectral split)**.** *For every smooth admissible h,*

$$E(h) \leq \frac{1}{58}N + \frac{9}{319}N_3. \tag{6}$$

Proof. Equation (4) yields

$$E(h_\ell) = \frac{1}{2(\lambda_\ell - 1)}\|A_{h_\ell}\|_2^2.$$

The coefficient equals $1/22$ at $\ell = 3$ and is at most $1/58$ for every odd $\ell \geq 5$. Orthogonality therefore gives

$$E(h) \leq \frac{1}{22}N_3 + \frac{1}{58}(N - N_3) = \frac{1}{58}N + \frac{9}{319}N_3.$$

□

3. The degree-three determinant inequality

Every $Y \in \mathcal{H}_3$ is the restriction to $\mathbb{S}^2$ of a homogeneous harmonic cubic

$$F(x) = T_{ijk}x_i x_j x_k,$$

where T is a real symmetric trace-free rank-three tensor: $T_{ijk} = T_{(ijk)}$ and $T_{iik} = 0$. Fix $u \in \mathbb{S}^2$ and define

$$P = I - u\otimes u, \qquad M(u)_{jk} = T_{ijk}u_i, \qquad v(u) = M(u)u, \qquad Y(u) = u^T M(u)u.$$

Also put

$$\tau = T_{ijk}T_{ijk}, \qquad B_{ij} = T_{ikl}T_{jkl}, \qquad B_0 = B - \frac{\tau}{3}I. \tag{7}$$

Lemma 3.1 (Spherical curvature tensor)**.** *For $C = A_Y = \nabla^2_{\mathbb{S}^2}Y + Y\,\mathrm{Id}$, regarded as a 3×3 symmetric matrix that vanishes on the normal direction u,*

$$C = 6PMP - 2YP. \tag{8}$$

Moreover,

$$\operatorname{tr} C = -10Y, \tag{9}$$

$$|C|^2 = 36\,|M|^2 - 72\,|v|^2 + 68Y^2, \tag{10}$$

$$4 \det C = 64Y^2 + 144\,|v|^2 - 72\,|M|^2 . \tag{11}$$

The determinant is taken on $T_u\mathbb{S}^2$.

Proof. For tangent vectors $\xi, \eta \in T_u\mathbb{S}^2$, the restriction formula for the Hessian of a homogeneous cubic gives

$$\nabla^2_{\mathbb{S}^2} Y(\xi,\eta) = D^2F(u)[\xi,\eta] - 3Y(u)\,\langle \xi,\eta\rangle\,.$$

Since $D^2F(u) = 6M(u)$, adding $Y\,\mathrm{Id}$ yields (8). Trace-freeness gives $\operatorname{tr} M = 0$ and $\operatorname{tr}(PMP) = -Y$, proving (9).

In an orthonormal basis whose first vector is u, write

$$M = \begin{pmatrix} Y & w^T \\ w & N \end{pmatrix}, \qquad v = \begin{pmatrix} Y \\ w \end{pmatrix}.$$

Then $|PMP|^2 = |M|^2 - 2\,|v|^2 + Y^2$. Expanding $|6PMP - 2YP|^2$ gives (10). Finally, for the two tangent eigenvalues of C,

$$4 \det C = 2\big((\operatorname{tr} C)^2 - \operatorname{tr}(C^2)\big),$$

and substitution of (9)–(10) yields (11). $\square$

We use the standard even moment formula

$$\int_{\mathbb{S}^2} u_{i_1}\cdots u_{i_{2m}}\,d\sigma = \frac{4\pi}{(2m+1)!!}\sum_{\mathcal{P}}\prod_{\{a,b\}\in\mathcal{P}} \delta_{i_a i_b}, \tag{12}$$

where the sum runs over all pairings of $\{1,\dots,2m\}$. Odd moments vanish. The quadratic consequences needed below are

$$\int_{\mathbb{S}^2} Y^2\,d\sigma = \frac{8\pi}{35}\tau, \qquad \int_{\mathbb{S}^2} |v|^2\,d\sigma = \frac{8\pi}{15}\tau, \qquad \int_{\mathbb{S}^2} |M|^2\,d\sigma = \frac{4\pi}{3}\tau. \tag{13}$$

Combining (10) with (13) gives

$$\|C\|_2^2 = \frac{176\pi}{7}\tau. \tag{14}$$

For the quartic calculation, define

$$J := \operatorname{tr}(B^2), \qquad \mathcal{K} := T_{abc}T_{ade}T_{bdf}T_{cef}.$$

The next relation is special to dimension three.

Lemma 3.2 (Quartic contraction relation)**.**

$$\mathcal{K} = \frac{1}{2}\tau^2 - J. \tag{15}$$

Proof. For each i, let $(A_i)_{jk} = T_{ijk}$. Each A_i is a symmetric traceless 3×3 matrix. One has

$$\tau = \sum_i \operatorname{tr}(A_i^2), \qquad J = \sum_{i,j}\big(\operatorname{tr}(A_iA_j)\big)^2, \qquad \sum_i A_i^2 = B.$$

For every traceless 3×3 matrix X, Cayley–Hamilton implies $\operatorname{tr}(X^4) = \frac{1}{2}(\operatorname{tr}(X^2))^2$. Apply this to $X = A + tD$ and compare the coefficient of t^2:

$$4\operatorname{tr}(A^2D^2) + 2\operatorname{tr}(ADAD) = \operatorname{tr}(A^2)\operatorname{tr}(D^2) + 2(\operatorname{tr}(AD))^2.$$

Set $A = A_i$, $D = A_j$, and sum over i, j. The first term on the left sums to $4J$, while

$$\sum_{i,j} \operatorname{tr}(A_i A_j A_i A_j) = T_{iab}T_{jbc}T_{icd}T_{jda} = \mathcal{K}$$

after relabeling dummy indices and using full symmetry of T. The right side sums to $\tau^2 + 2J$. Thus $4J + 2\mathcal{K} = \tau^2 + 2J$, which is (15). ☐

The pairing expansion from (12), together with Lemma 3.2, yields the six moment identities in Table 1. Appendix A gives the complete contraction count.

TABLE 1. Quartic moments after reducing $\mathcal{K}$ by (15).

Integrand	Integral divided by π
Y^4	$\frac{16}{715}\tau^2 + \frac{96}{5005}J$
$Y^2 \lvert v\rvert^2$	$\frac{16}{495}\tau^2 + \frac{32}{1155}J$
$Y^2 \lvert M\rvert^2$	$\frac{8}{315}\tau^2 + \frac{16}{105}J$
$\lvert v\rvert^4$	$\frac{16}{315}\tau^2 + \frac{32}{315}J$
$\lvert v\rvert^2 \lvert M\rvert^2$	$\frac{8}{105}\tau^2 + \frac{32}{105}J$
$\lvert M\rvert^4$	$\frac{4}{15}\tau^2 + \frac{8}{15}J$

Proposition 3.3 (Exact cubic determinant identity). *For every $Y \in \mathcal{H}_3$ and $C = A_Y$,*

$$\|4 \det C\|^2_{L^2(\mathbb{S}^2)} = \frac{\pi}{1001}\left(555264\tau^2 - 2265600\,|B_0|^2\right). \tag{16}$$

Consequently,

$$\|4 \det C\|^2_{L^2(\mathbb{S}^2)} \le \frac{15183}{17303\pi}\,\|C\|_2^4. \tag{17}$$

Equality in (17) holds exactly when $B_0 = 0$; this occurs, for example, for $Y(x, y, z) = xyz$ up to scaling and rotation.

Proof. Put $a = Y^2$, $b = |v|^2$, and $c = |M|^2$. By (11),

$$(4 \det C)^2 = 4096a^2 + 18432ab - 9216ac + 20736b^2 - 20736bc + 5184c^2.$$

Substitution of Table 1 gives

$$\int_{\mathbb{S}^2} (4 \det C)^2\, d\sigma = \frac{\pi}{1001}\left(1310464\tau^2 - 2265600J\right).$$

Since $\operatorname{tr} B = \tau$ and $B_0 = B-(\tau/3)I$, $J = |B_0|^2 + \tau^2/3$, which proves (16). Dropping the nonpositive B_0 term and eliminating τ using (14) yields

$$\frac{555264\pi/1001}{(176\pi/7)^2} = \frac{15183}{17303\pi},$$

proving (17). Equality is equivalent to $B_0 = 0$. For $Y = xyz$, the six components T_{123} and their permutations are $1/6$, which gives $\tau = 1/6$ and $B = (1/18)I = (\tau/3)I$. ☐

4. The degree-three curvature budget

Set

$$s := \sqrt{\frac{15183}{17303}}.$$

For a self-adjoint endomorphism M of a two-dimensional Euclidean space, write $\|M\|_{\mathrm{op}}$ for the operator norm and $\|M\|_*$ for the nuclear norm.

Lemma 4.1 (Integrated nuclear-norm bound)**.** *For $C = A_Y$ with $Y \in \mathcal{H}_3$,*

$$\int_{\mathbb{S}^2} \|C\|_*^2 \, d\sigma \le (1+s)\, \|C\|_2^2\,, \tag{18}$$

$$\left(\int_{\mathbb{S}^2} \|C\|_* \, d\sigma\right)^2 \le 4\pi(1+s)\, \|C\|_2^2\,. \tag{19}$$

Proof. If μ_1, μ_2 are the tangent eigenvalues of C, then

$$\|C\|_*^2 = (|\mu_1| + |\mu_2|)^2 = |C|^2 + 2\,|\det C|\,.$$

Writing $q = 4 \det C$ and integrating,

$$\int_{\mathbb{S}^2} \|C\|_*^2 \, d\sigma = \|C\|_2^2 + \frac{1}{2}\, \|q\|_{L^1}\,.$$

Cauchy–Schwarz and (17) give

$$\|q\|_{L^1} \le 2\sqrt{\pi}\, \|q\|_{L^2} \le 2s\, \|C\|_2^2\,,$$

which proves (18). A second Cauchy–Schwarz inequality on the sphere of area 4π gives (19). □

Proposition 4.2 (Cubic curvature budget)**.** *For the degree-three component $C = A_{h_3}$ of any smooth admissible support function,*

$$N_3 = \|C\|_2^2 \le \pi d^2 (1+s). \tag{20}$$

Proof. For self-adjoint tangent endomorphisms A, M,

$$|\mathrm{tr}(AM)| \le \|A\|_{\mathrm{op}}\, \|M\|_*\,.$$

By (2), $\|A_h(u)\|_{\mathrm{op}} \le d/2$. The projection identity (5) therefore gives

$$N_3 = \int_{\mathbb{S}^2} \mathrm{tr}(A_h C)\, d\sigma \le \frac{d}{2} \int_{\mathbb{S}^2} \|C\|_* \, d\sigma.$$

If $N_3 = 0$ there is nothing to prove. Otherwise square, apply (19), and divide by N_3:

$$N_3 \le 4\pi \left(\frac{d}{2}\right)^2 (1+s) = \pi d^2 (1+s).$$

□

5. The volume bound

Theorem 5.1. *Every convex body $K \subset \mathbb{R}^3$ of constant width d satisfies*

$$\mathrm{Vol}(K) \ge \frac{\pi}{1914} \left(259 - 27\sqrt{\frac{15183}{17303}}\right) d^3 = 0.3836027047090677\ldots\, d^3. \tag{21}$$

Proof. First suppose the support function is smooth. By (2), the two eigenvalues of $A_h(u)$ lie in $[-d/2, d/2]$, so

$$N = \|A_h\|_2^2 \le 2\pi d^2. \tag{22}$$

Combining Proposition 2.3, Proposition 4.2, and (22),

$$\begin{aligned} E(h) &\le \frac{1}{58}N + \frac{9}{319}N_3 \\ &\le \frac{2\pi d^2}{58} + \frac{9}{319}\pi d^2(1+s) \\ &= \frac{\pi d^2}{319}(20+9s). \end{aligned}$$

Insert this in (3):

$$\mathrm{Vol}(K) \ge \pi d^3\left(\frac{1}{6} - \frac{20+9s}{638}\right) = \frac{\pi d^3}{1914}(259-27s).$$

This proves (21) in the smooth class.

For an arbitrary constant-width body, use the smooth constant-width approximation and Hausdorff continuity of volume recorded by Nishioka [6, Lemma 2], ultimately based on [5, Theorem 8.5.1]. Apply the smooth estimate to the approximating bodies and pass to the limit. □

Corollary 5.2. *The coefficient in (21) is strictly larger than $4\pi/33$. More precisely, with $s = \sqrt{15183/17303}$,*

$$\frac{\pi}{1914}(259-27s) - \frac{4\pi}{33} = \frac{9\pi}{638}(1-s) = 0.00280359518303217\ldots > 0.$$

6. Reproducibility and disclosure

The accompanying repository separates the mathematical derivation from the computational checks. The exact Python/SymPy scripts:

- construct a general seven-parameter symmetric trace-free cubic tensor;
- derive (14) and (16) by exact polynomial integration on $\mathbb{S}^2$;
- check (11) independently modulo $x^2+y^2+z^2-1$;
- enumerate all pairings behind Appendix A; and
- reproduce the exact and decimal constants in Theorem 5.1.

Independent base-R scripts repeat the pairing count and numerical tensor stress tests. These computations are evidence against transcription and normalization errors; they do not substitute for independent mathematical review.

Declaration of AI assistance. AI assistance was used extensively to explore proof strategies, carry out and cross-check symbolic calculations, draft exposition, and prepare reproducibility code. The named author is responsible for the decision to release this draft, for accurately describing its provenance and verification status, and for maintaining corrections. No AI system is listed as an author. Independent expert verification of the complete proof has not yet occurred.

Appendix A. Contraction counts for the quartic moments

Represent the four copies of T by four labeled vertices, each with three index slots. A contraction within a single tensor is a loop and vanishes because T is trace-free. Every surviving complete contraction is one of three graph types:

(1) D: two disjoint vertex pairs joined by three parallel edges, giving τ^2;
(2) J: two opposite doubled edges and two connecting single edges, giving $J = \mathrm{tr}(B^2)$;
(3) $\mathcal{K}$: the simple complete graph on four vertices, giving the tetrahedral contraction $\mathcal{K}$.

The relation $\mathcal{K} = \tau^2/2 - J$ then reduces every quartic moment to τ^2 and J.

Some scalar contractions are already fixed before applying the sphere-moment formula. Table 2 records the remaining pairing counts. The last two columns result after eliminating $\mathcal{K}$.

TABLE 2. Exact pairing counts.

Integrand	n_D	n_J	$n_{\mathcal{K}}$	n_{τ^2}	n'_J
Y^4	108	1944	1296	756	648
$Y^2\,\lvert v\rvert^2$	12	216	144	84	72
$Y^2\,\lvert M\rvert^2$	6	36	0	6	36
$\lvert v\rvert^4$	4	40	16	12	24
$\lvert v\rvert^2\,\lvert M\rvert^2$	2	8	0	2	8
$\lvert M\rvert^4$	1	2	0	1	2

For example, in Y^4 the disconnected type contributes $3(3!)^2 = 108$ pairings, the tetrahedral type contributes $(3!)^4 = 1296$, and the six labeled J graphs contribute $6 \cdot 3^4(2!)^2 = 1944$. The other rows follow by the same slot-counting argument with the scalar contractions in $|v|^2$ and $|M|^2$ fixed. The repository script `python/verify_pairing_counts.py` exhaustively enumerates the pairings and reproduces the table exactly; the base-R script `R/verify_pairing_counts.R` is an independent implementation.

Multiplication by $4\pi/(2m+1)!!$, where $2m$ is the number of remaining sphere-coordinate factors, gives Table 1.

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